

Project 5 — Finance Market Analysis

Dataset note: This is simulated financial data for portfolio practice. It is not real market data or investment advice.

Executive Summary

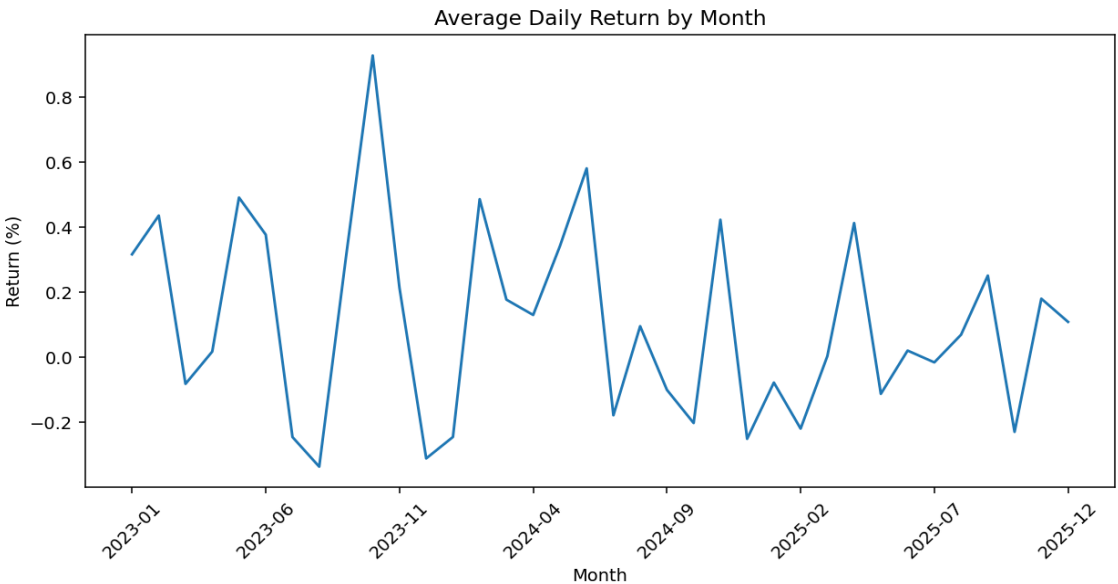
The cleaned dataset contains 1,500 synthetic observations. Average daily return is 0.099% and estimated annualized volatility is 28.13%. 53.2% of observations have positive daily returns. Sector and sentiment analyses demonstrate descriptive financial analytics while avoiding predictive claims.

Data Quality

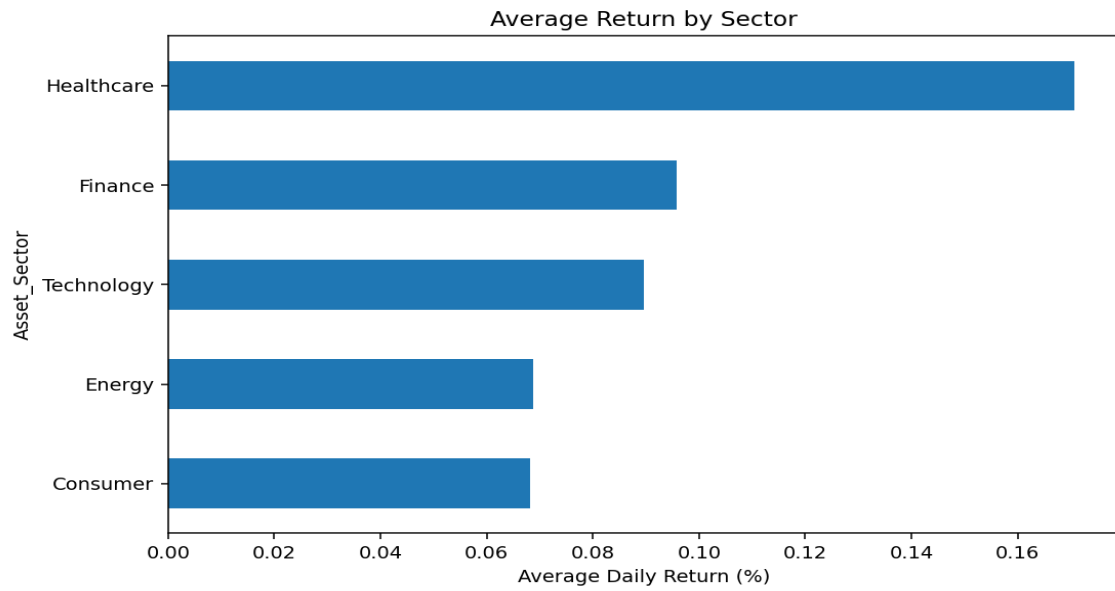
Metric	Result
Raw rows	1,502
Clean rows	1,500
Duplicates removed	2
Missing values	Median imputation

Sector Summary

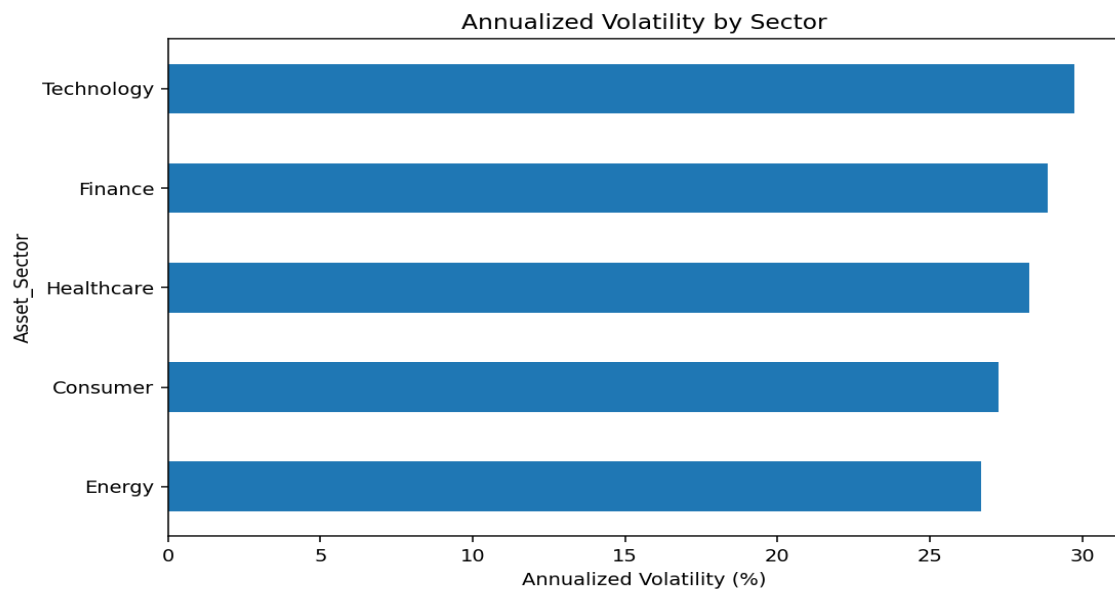
Sector	Avg Return %	Volatility %	Avg Volume
Healthcare	0.171	28.27	565,594
Finance	0.096	28.88	559,331
Technology	0.090	29.74	539,272
Energy	0.069	26.68	532,242
Consumer	0.068	27.26	537,527



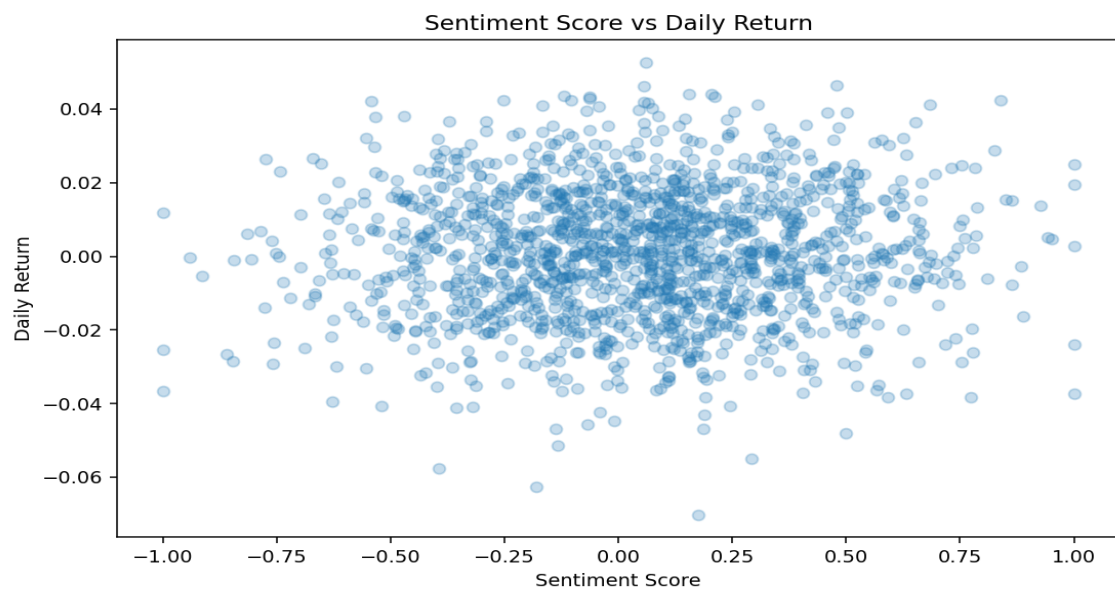
Average daily return by month



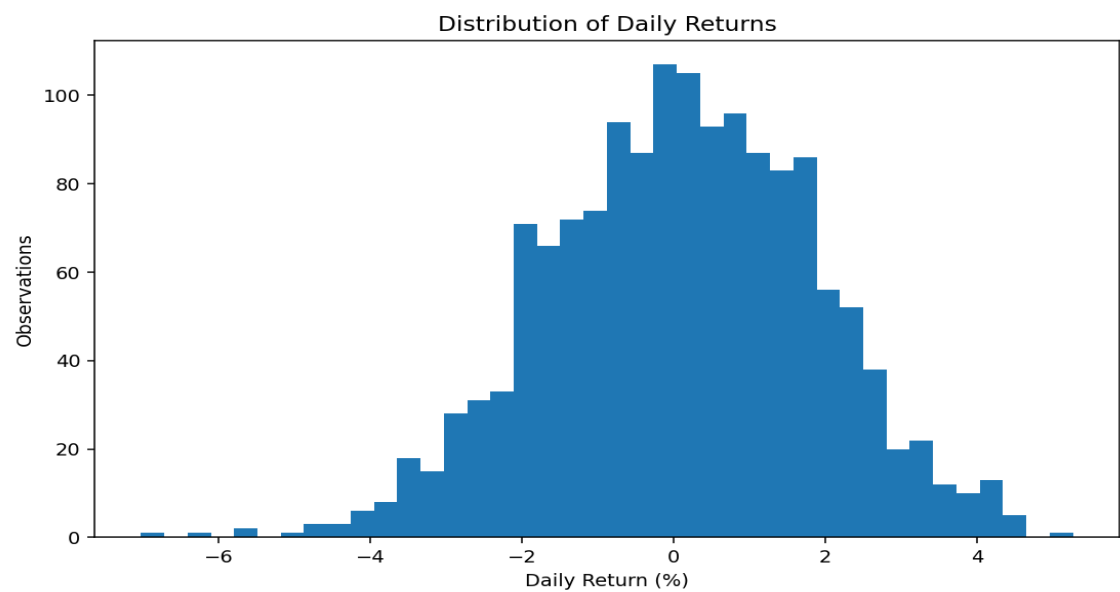
Average return by sector



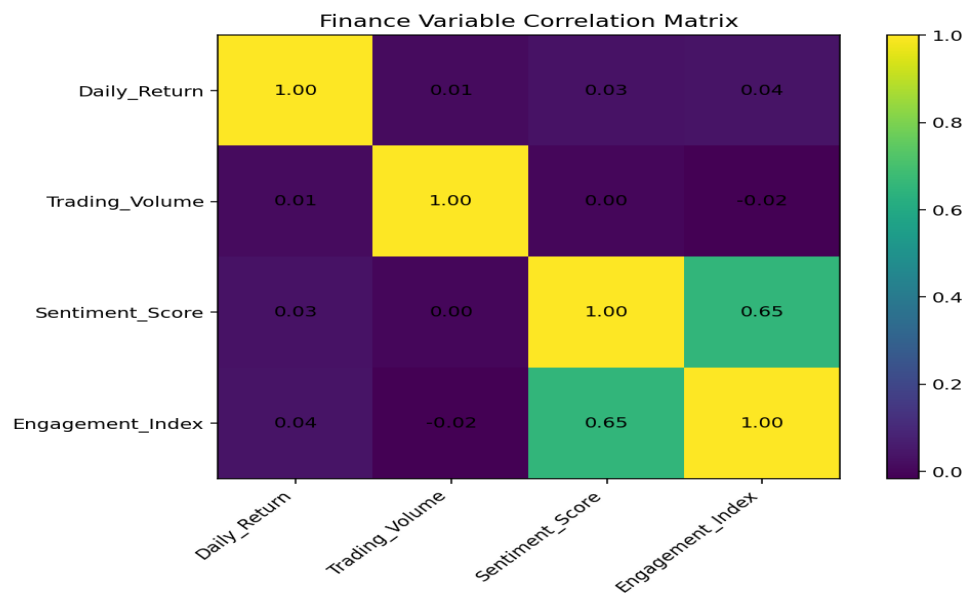
Annualized volatility by sector



Sentiment score versus daily return



Distribution of daily returns



Finance-variable correlation matrix

Interpretation and Limitations

Return and volatility are descriptive measures from simulated observations. Correlation can identify relationships worth investigating but cannot establish causation. The dataset is synthetic, so no result should be generalized to real markets or used to make investment decisions.